ME 255
Strength of Materials I

Lecture 2: Tension In Structural Members



Lecture Objectives

- Stress and strain in structural members
- Thermal Deformation
- Variable Load
- Tensile Test

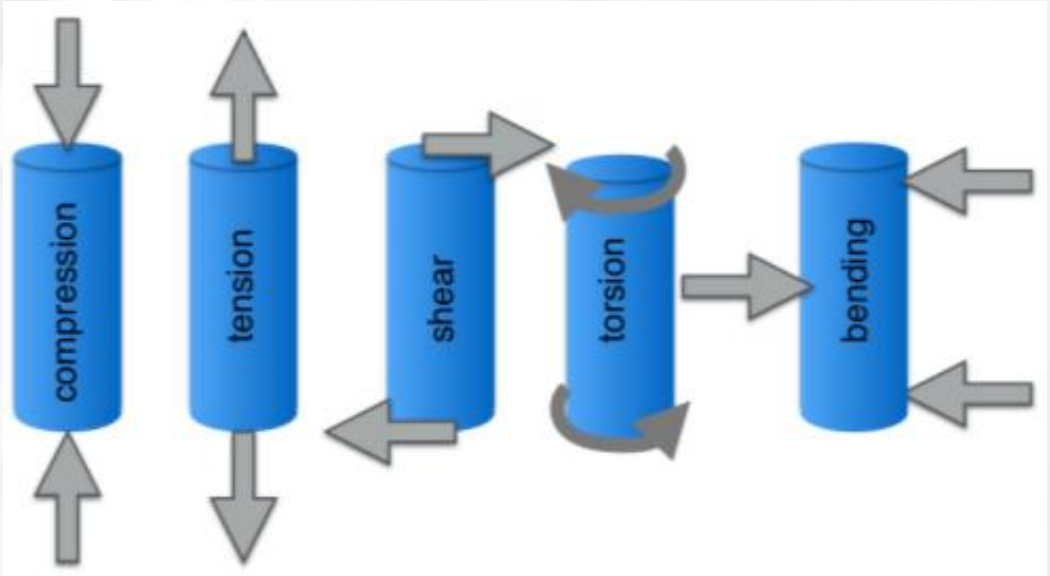
Classification of Stresses and Strains

- Stresses can be classified based on two conditions.

1. The type of loading in the machine member or the structure.

Therefore, the following types of stresses can be identified with the corresponding type of loading;

a) Axial stress due to axial loading
b) Bending stress due to bending in beams
c) Shearing stress due to torsion in shafts
d) Shearing stress due to transverse or shear load
e) Hood/Transverse stress in pressure vessels
f) Axial stress in pressure vessels



Classification of Stresses and Strains

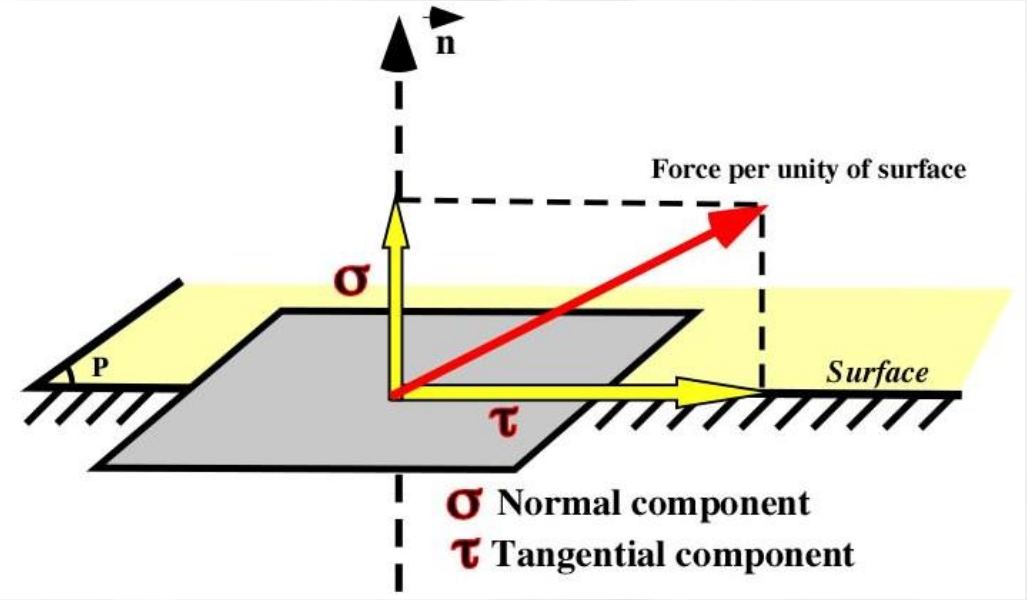
2. ***How the force-couple acts on the surface of the machine or structure***.

Thus, we have:

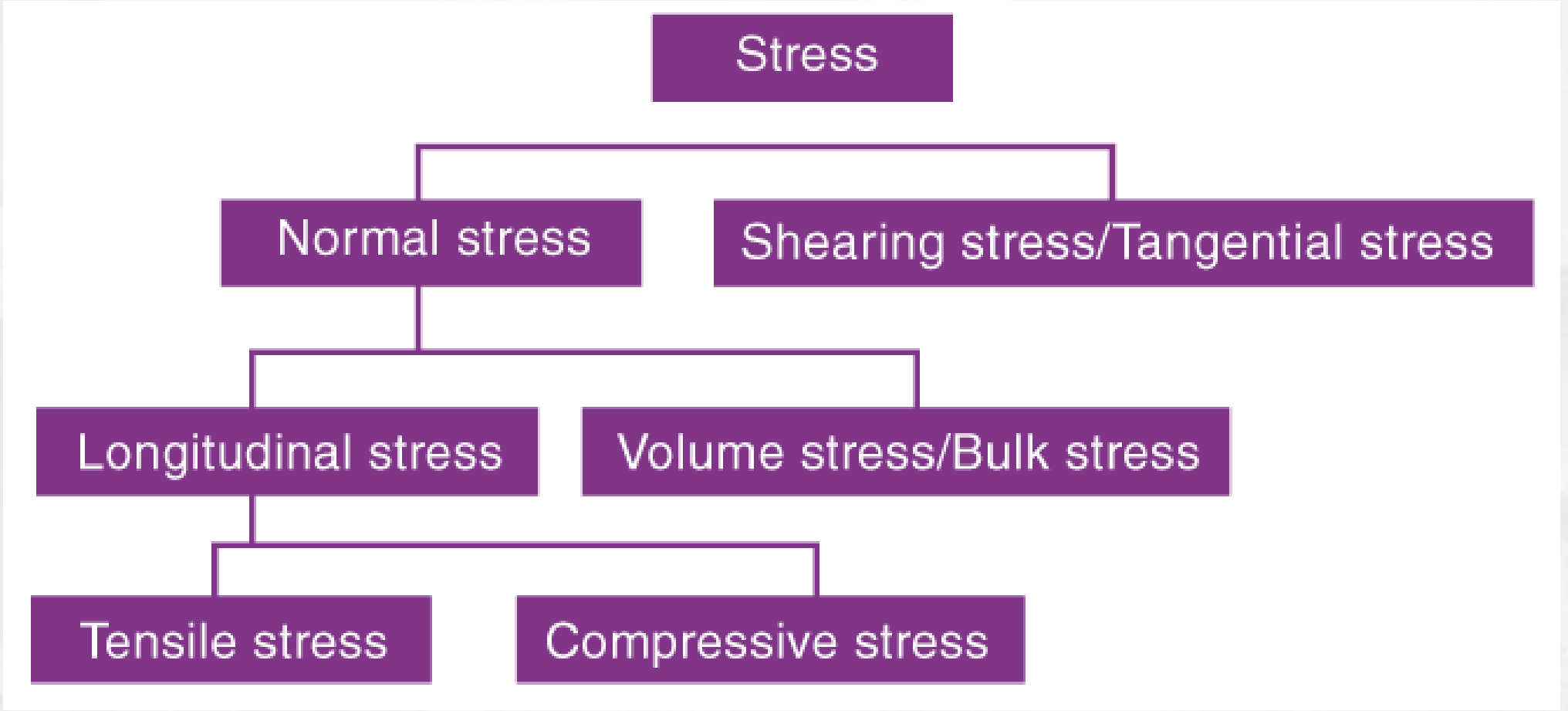
a. Normal stress acting perpendicular to the surface area

b. Shearing stress acting tangential to the surface area

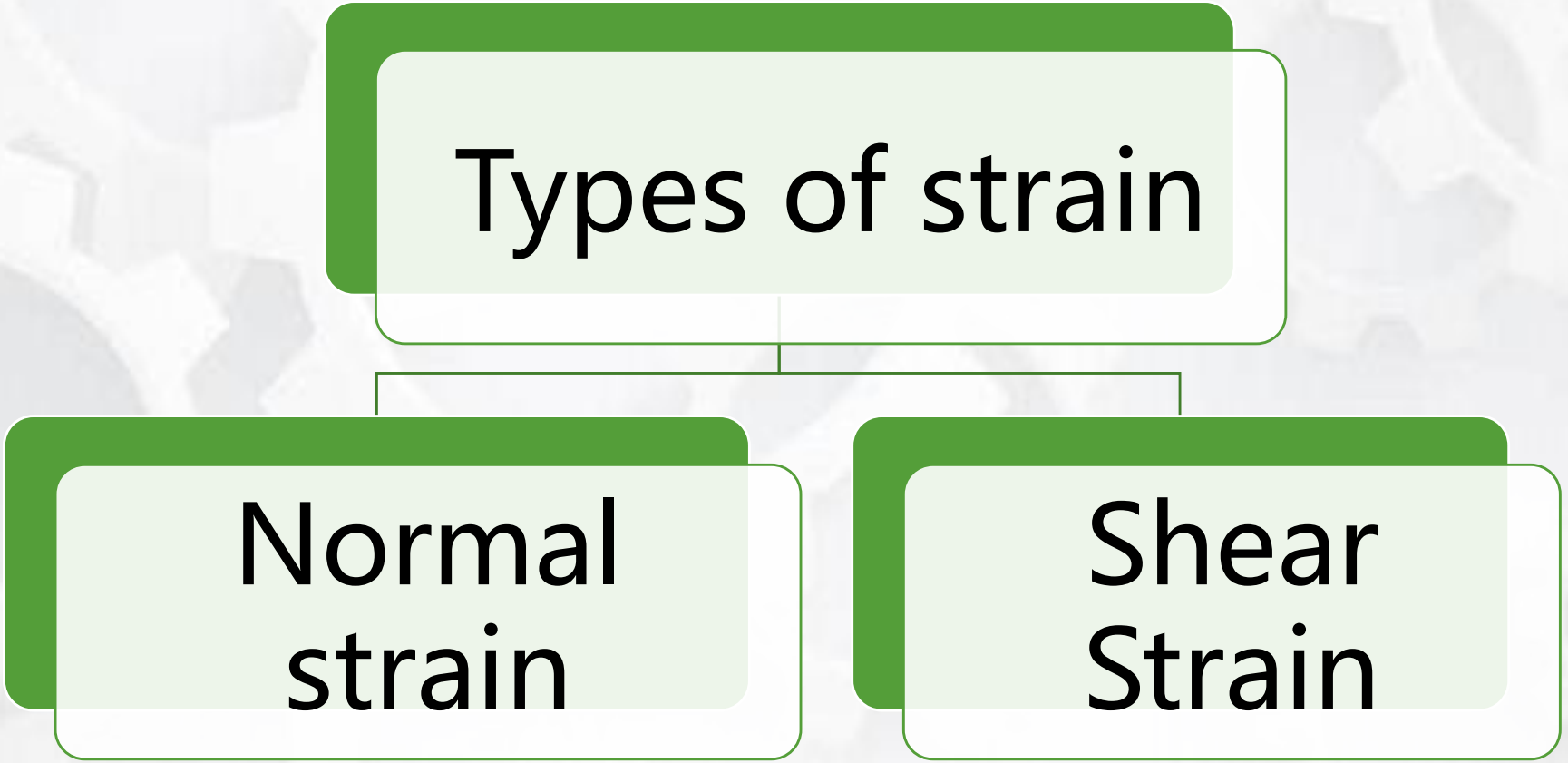
c. Bearing stress acting on connecting members (bolts. rivets, pins)



Classification of Stresses and Strains



Classification of Stresses and Strains



Stress and Strain within the Elastic Limits

- By considering engineering structures as deformable and analyzing the deformations in their various members, it will be possible for as to

  i. compute forces which are statically indeterminate, indeterminate within the framework of statics.
  ii. determine the actual distribution of stresses within a member. It is thus, necessary to analyze the deformation which take place in that member.

- We will consider the deformations of a structural member such as a rod, bar, or plate under axial loading

Stress and Strain within the Elastic Limits

- If a force P is applied to a member as shown in Fig. 1 (a),

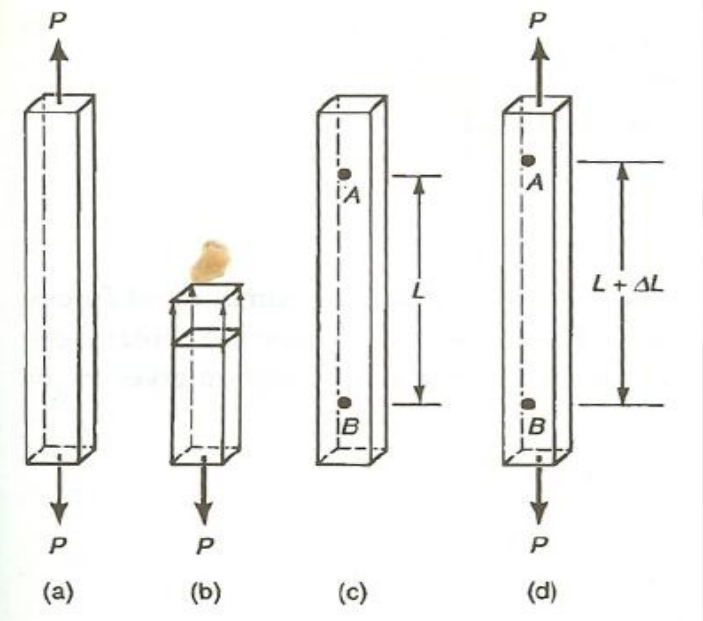


Fig. 1

- The force per unit area is called the stress and is given the symbol **σ**.
- Thus, $\sigma = P/A$
- The ultimate stress is therefore $\sigma_{ult} = \frac{P_{ult}}{A}$
- When the specimen is subjected to load, P very beyond the ultimate, necking of the material occurs and then rapture.
- Assuming the area of the neck is A_n, the yielding stress is given as

$$\sigma_y = \frac{P_y}{A_n}$$

Example 2-1

A tensile is carried out on a bar of mild steel of diameter 2 cm. The bar yield under a load of 80 kN. It reaches a maximum load of 150 kN, and breaks finally at a load of 70 kN. Find the following:

a. the tensile stress at the yield point
b. the ultimate tensile stress
c. the average stress at the breaking point if the diameter of the fractured neck is 1 cm.

Solution

The initial cross-section of the bar is

$$A_i = \frac{\pi}{4} d_i^2 = \frac{\pi}{4} (0.02)^2 = 0.000314 m^2$$

Example 2-1 Continues

a. the average tensile stress at yielding point

$$\sigma_y = \frac{P_y}{A_i} = \frac{80000}{0.000314} = 254MPa$$

b. the ultimate stress is the stress at the maximum load,

$$\sigma_u = \frac{p_{\max}}{A_i} = \frac{150000}{0.000314} = 477MPa$$

c. the cross-sectional area in the fractured neck is

$$A_f = \frac{\pi}{4} d_f^2 = \frac{\pi}{4} (0.01)^2 = 0.0000785m^2$$

The average stress at the breaking point is

$$\sigma_f = \frac{P_f}{A_f} = \frac{70000}{0.0000785} = 892MPa$$

Deformation under Tension & Compression

- A quantity measuring the intensity of deformation with respect to the original length L is the strain **ε**, defined as:

$$\varepsilon = \Delta L / L = \delta / L$$

where **ΔL** is denoted as δ.

- The relationship between stress and strain is determined experimentally.

- A typical plot of stress versus strain is shown in Figure 2.

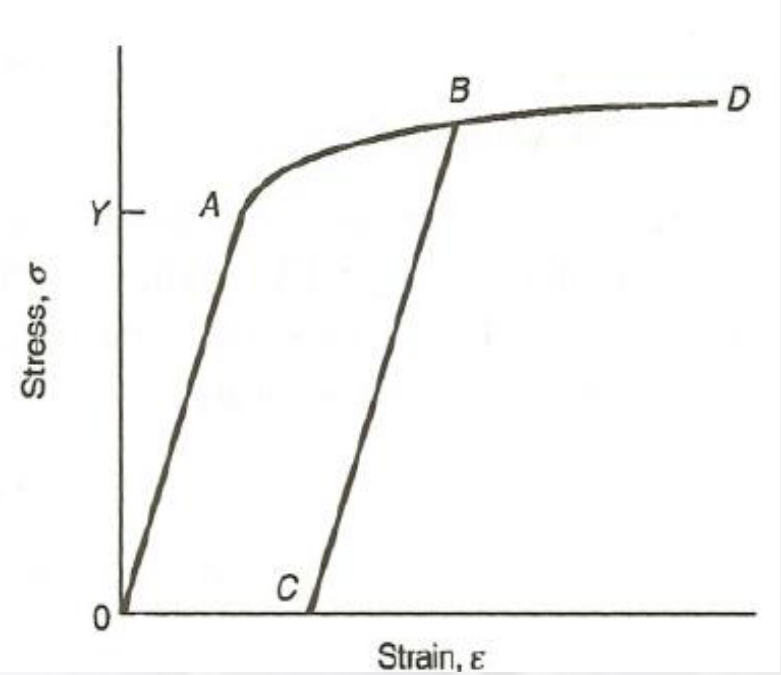


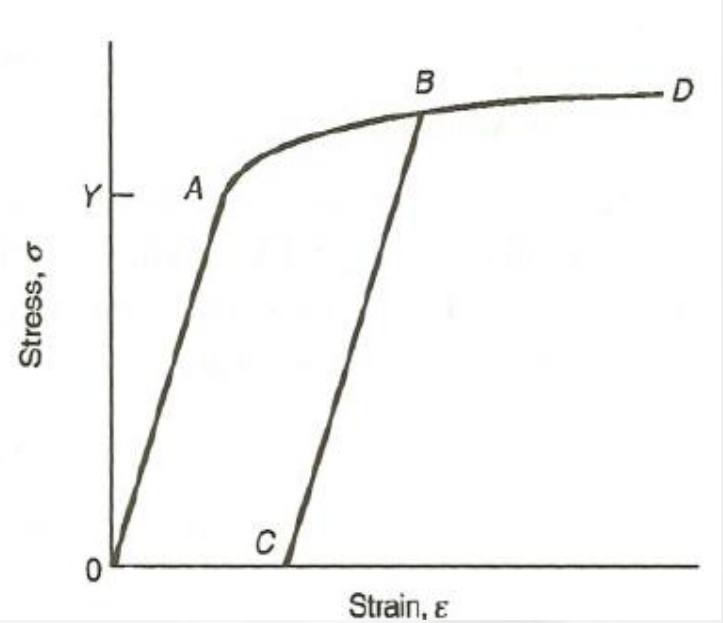
Fig. 2

Modulus of Elasticity

- The region of greatest concern is that below the yield point.

- The slope of the line between O and A is called the modulus of elasticity and is given the symbol **E**, so

$$\boldsymbol{E} = \frac{Stress\ (\sigma)}{Strain\ (\varepsilon)}$$



- The relation of the applied force in a member to its axial deformation can be found by inserting the definitions of the stress and the axial strain into Hooke's Law [Eq.(2-5)],which gives

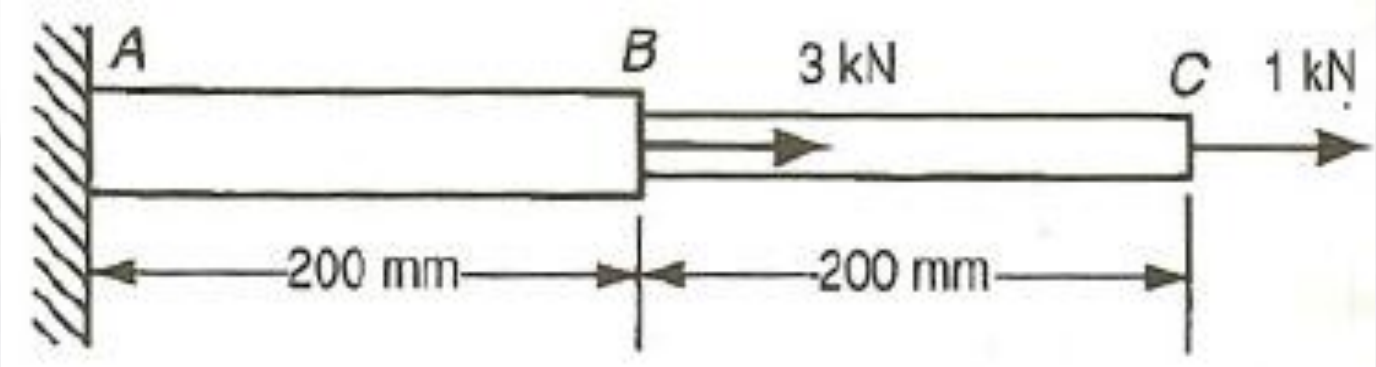
$$\delta = \frac{PL}{AE}$$

In the examples that follow, wherever it is appropriate, the three steps of ***Equilibrium***, ***Force-Deformation***, and ***Compatibility*** will be explicitly stated.

Example 2-2

The steel rod shown below is fixed to a wall at its left end. It has two applied forces. The 3 kN force is applied at the point B and the 1 kN force is applied at the point C. The area of the rod between A and B is 1000 mm^2, and the area of the rod between B and C is 500 mm^2. Take E= 210 GPa. Find:

a. the stress in each section of the rod and
b. the horizontal displacement at the points B and C

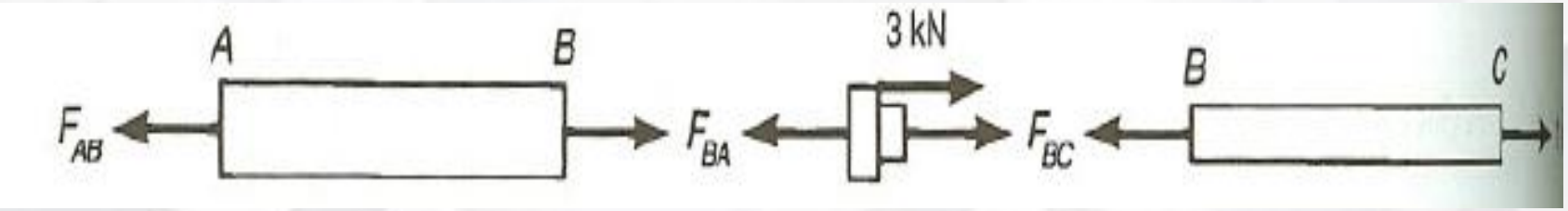


Example 2-2 Continues

Solution

I: Equilibrium

Draw free-body diagrams for each section of the rod.



From a summation of forces on the member BC, F_{BC}=1 kN.
Summing forces on the centre free-body diagram, F_{BA} = 3+1=4 kN

Summing forces on the left free-body diagram gives $F_{AB} = F_{BA}$ = 4 kN.

The stresses then are:

$$\sigma_{AB} = 4kN/1000mm^2 = 4MPa$$

$$\sigma_{BC} = 1kN/500mm^2 = 2MPa$$

Example 2-2 Continues

II: Force-Deformation

$$\delta_{AB} = \left(\frac{PL}{AE}\right)_{AB} = \left[\frac{(4kN)(200mm)}{(1000mm^2)(210GPa)}\right]$$

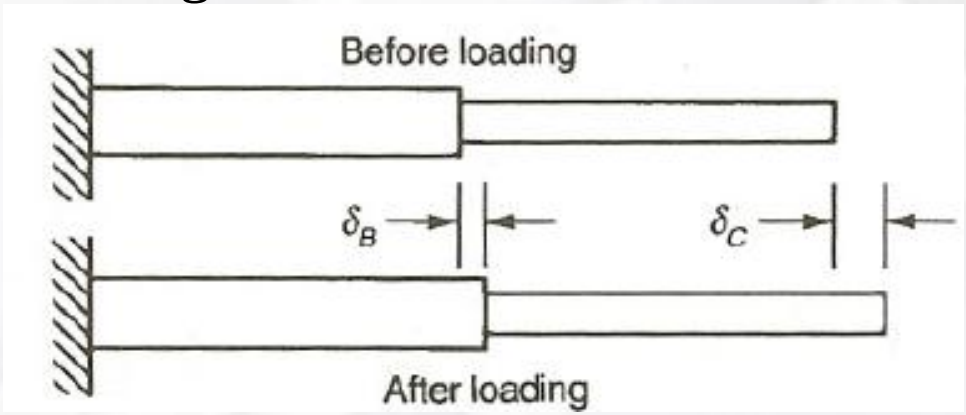
$$\delta_{AB} = 3.81x10^{-3}\,mm$$

$$\delta_{BC} = \left(\frac{PL}{AE}\right)_{BC} = \left[\frac{(1kN)(200mm)}{(500mm^2)(210GPa)}\right]$$

$$\delta_{BC} = 1.91x10^{-3}\,mm$$

III: Compatibility

Draw the body before loading and after loading



It is then obvious that

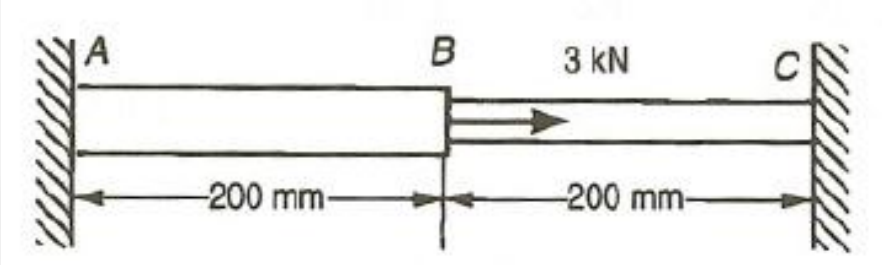
$$\delta_B = \delta_{AB} = 3.81X10^{-3}\,mm$$

$$\delta_C = \delta_{AB} + \delta_{BC} = \left[3.81+1.91\right]x10^{-3} = 5.71x10^{-3}\ mm$$

Example 2-3

Consider the same steel rod as in Example 2-2 except that now the right end is fixed to a wall as well as the left (Figure below). It is assumed that the rod is built into the walls before the load is applied. Find:

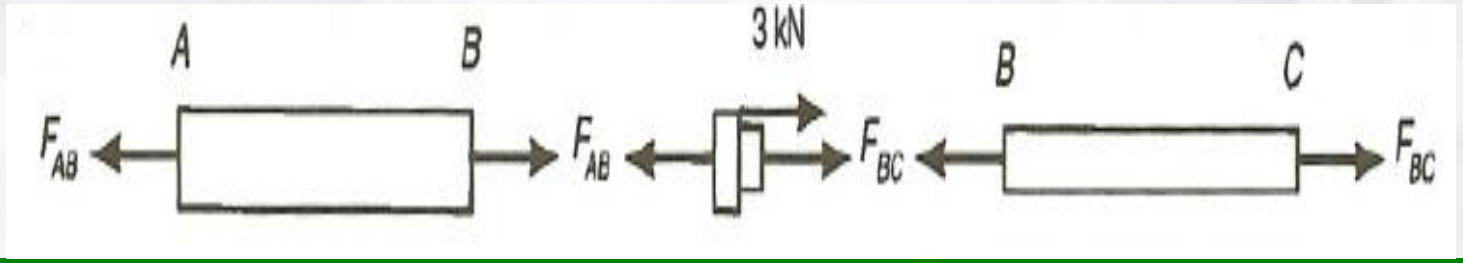
a. the stress in each section of the rod, and
b. the horizontal displacement at point B.



Solution

I: Equilibrium

Draw free-body diagrams for each section of the rod.



Example 2-3 Continues

Summing forces in the horizontal direction on the centre free-body diagram

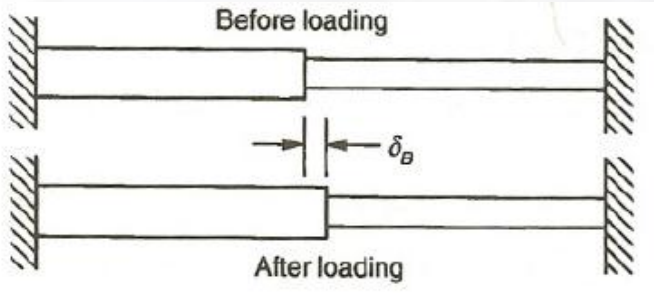
$$-F_{AB} + F_{BC} + 3 = 0 \qquad \textbf{a}$$

II: Force-Deformation

$$\delta_{AB} = \left(\frac{PL}{AE}\right)_{AB} = \frac{F_{AB}L_{AB}}{A_{AB}E} \qquad \textbf{b}$$

$$\delta_{BC} = \left(\frac{PL}{AE}\right)_{BC} = \frac{F_{BC}L_{BC}}{A_{BC}E} \qquad \textbf{c}$$

III: Compatibility



$$\delta_C = \delta_{AB} + \delta_{BC}$$

But $\delta_C = 0$

$$\therefore \delta_{AB} = -\delta_{BC} \Rightarrow \frac{F_{AB}L_{AB}}{A_{AB}E} = \frac{F_{BC}L_{BC}}{A_{BC}E}$$

Example 2-3 Continues

Since $L_{AB} = L_{BC}$ and $A_{AB} = 2A_{BC}$, then

$$F_{AB} = -2F_{BC}$$

Inserting this relation into the equilibrium equation (Eq. a),

$$\Rightarrow -F_{AB} + F_{BC} + 3 = 2F_{BC} + F_{BC} + 3 = 0$$

$$\Rightarrow F_{BC} = -1kN \quad \text{and} \quad F_{AB} = 2kN$$

The stresses are $\sigma_{AB} = 2kN/1000\,mm^2 = 2MPa$

$$\sigma_{BC} = -1kN/500\,mm^2 = -2MPa$$

Example 2-3 Continues

The displacement at B is given by

$$\delta_B = \delta_{AB} = \left[\left(\frac{PL}{AE}\right)_{AB}\right] = \left[\frac{F_{AB}L_{AB}}{A_{AB}E}\right]$$

$$\delta_B = \left[\frac{(2)(200)}{(1000)(210)}\right] mm = 19.5x10^{-3}\ mm$$

Poisson's Ratio

- The ratio of the magnitude of the lateral strain to the magnitude of the longitudinal strain is called Poisson's Ratio, υ

$$\upsilon = -\frac{Lateral\ strain}{Longitudinal\ strain}$$

- Poisson's Ratio is a dimensionless material property that never exceeds 0.5.
- Typical values for steel, aluminum, and copper are 0.30, 0.33 and 0.34, respectively.

Example 2-4

A circular aluminum rod 10 mm in diameter is loaded with an axial force of 2 kN. What is the decrease in diameter of the rod? Take E = 70 GPa and $\upsilon = 0.33$.

Solution

The stress is $\sigma = P/A = (2)/(\pi 5^2) = 25.5 MPa$

The longitudinal strain is $\varepsilon_{lon} = \sigma/E = 25.5/70000 = 0.000364$

The lateral strain is $\varepsilon_{lat} = -\upsilon\varepsilon_{lon} = -(0.33)(0.000364) = -0.00012$

The decrease in diameter is then

$$\Delta d = -d\,\varepsilon_{lat} = -(10)(-0.00012) = 0.0012\ mm$$

Thermal Deformations

- When a material is heated, expansion forces are created.
- If it is free to expand, the thermal strain is given by

$$\varepsilon_t = \alpha(t - t_0)$$

- Where ε_t is thermal strain, α is linear thermal expansion coefficient, t is initial temperature and t_o is final temperature.
- For problems where the load is purely axial, this becomes

$$\varepsilon_T = \frac{\sigma}{A} + \alpha(t - t_0)$$

- The deformation δ is found by multiplying the strain by the length, L

$$\delta = \frac{PL}{AE} + \alpha L(t - t_0)$$

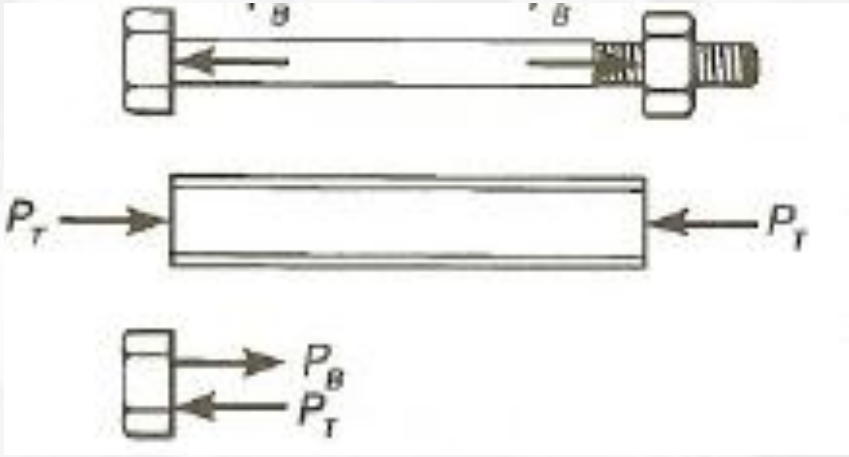
Example 2-5

The steel bolt is put through the aluminum tube as shown in Figure below. The nut is made just tight. The entire assembly is then raised in temperature by 60^0C. Because aluminum expands more than steel, the bolt will be put in tension and the tube in compression. Find the force in the bolt and the tube. For the steel bolt, take E = 210 GPa, α_T = 12E-06, A_T=32. For the aluminium tube, take E= 69GPa, α_B = 23E-06 and A_B=64



I: Equilibrium

Draw free-body diagrams for each section of the rod.



From equilibrium of the head of the bolt, it can be seen that $P_B = P_T$.

Example 2-5 Continues

II: Force-Deformation

$$\delta_B = \left(\frac{PL}{A_B E_B}\right) + \alpha_B L(t - t_0)$$

$$\delta_T = -\left(\frac{PL}{A_T E_T}\right) + \alpha_T L(t - t_0)$$

The minus sign in the second expression occurs because the tube is in compression

III: Compatibility

The tube and the bolt must both expand the same amount, therefore,

$$\delta_B = \delta_T = \left[\begin{array}{l} \frac{P(100)}{(32)(210E+9)} + \\ (12E-06)(100)(60) \end{array}\right]$$

$$\delta_B = \left[\begin{array}{l} \frac{P(100)}{(64)(69E+9)} + \\ (23E-06)(100)(60) \end{array}\right]$$

Solving for P gives P = 1.759 kN

Variable Load

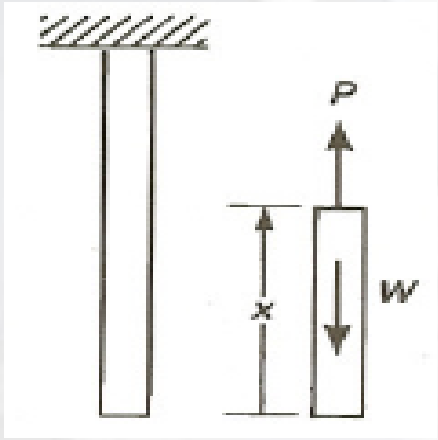
- For variable load, Eq. $\delta = \frac{PL}{AE}$ holds only over an infinitesimally small length L= dx.

- Eq. then becomes $$d\delta = \frac{P(x)}{AE}dx$$

- or equivalently $$\delta = \int_0^L \frac{P(x)}{AE}dx$$

Example 2-6

An aluminum rod is hanging from one end. The rod is 1m long and has a square cross-section 20 mm by 20 mm. Find the total extension of the rod resulting from its own weight. Take E = 70 GPa and the unit weight γ =27 kN/m3.



Solution

I: Equilibrium

Draw a free-body diagram.

The weight of the section shown in Figure E2-6 is

$$W = \gamma V = \gamma A x = P$$

which clearly yields P as a function of x, and Eq.(2-11) gives

$$\delta = \int_0^L \frac{\gamma A x}{AE} dx = \frac{\gamma}{E} \int_0^L x dx = \frac{\gamma L^2}{2E}$$

$$\delta = \frac{(27E + 3)(1)^2}{(2)(70E + 9)} = 0.1929 \mu m$$

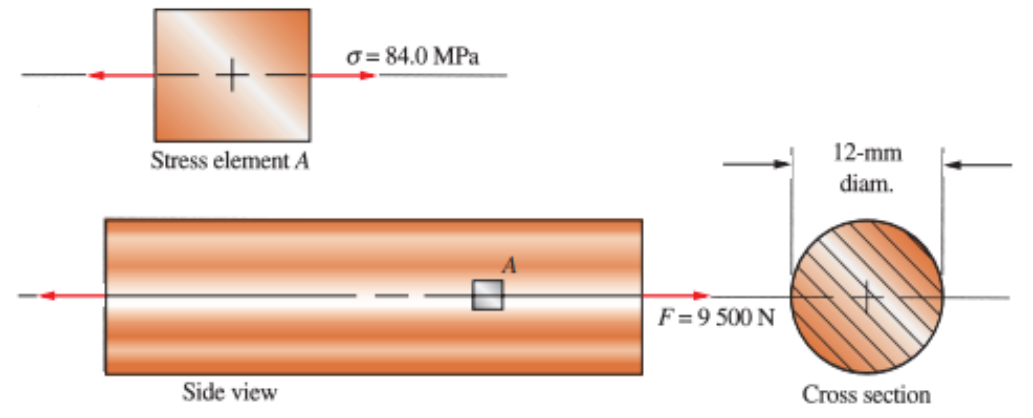
Trial Examples

1. A tensile force of 9500 N is applied to a 12-mm-diameter round bar, as shown in Fig. below. Compute the direct tensile stress in the bar.

Ans: 84 Mpa.

2. For the round bar subjected to the tensile load shown in (1), compute the total deformation if the original length of the bar is 3600 mm. The bar is made from a steel having a modulus of elasticity of 207 GPa.

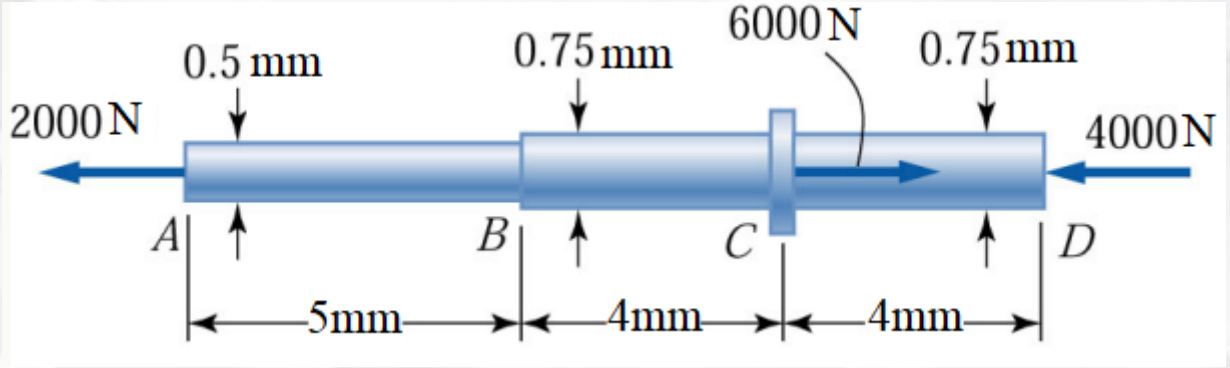
Ans: 1.46 mm



Trial Example

The propeller shaft ABCD carries the axial loads shown in Fig. below. Determine the change in the length of the shaft caused by these loads. Use $E = 29 \times 10^6$ Pa.

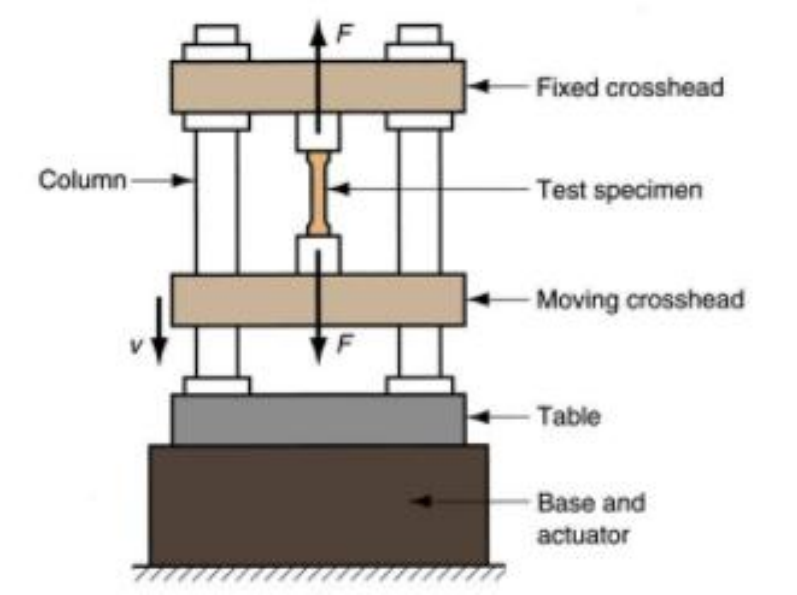
Ans: 0.01358 mm



Tensile Test

The tensile test measures the resistance of a material to a static or slowly changing force.

The test set-up is as shown in Fig. 2-3.



The stress $\sigma = F / A_0$

The strain, $\varepsilon = (l - l_0) / l_0$

where

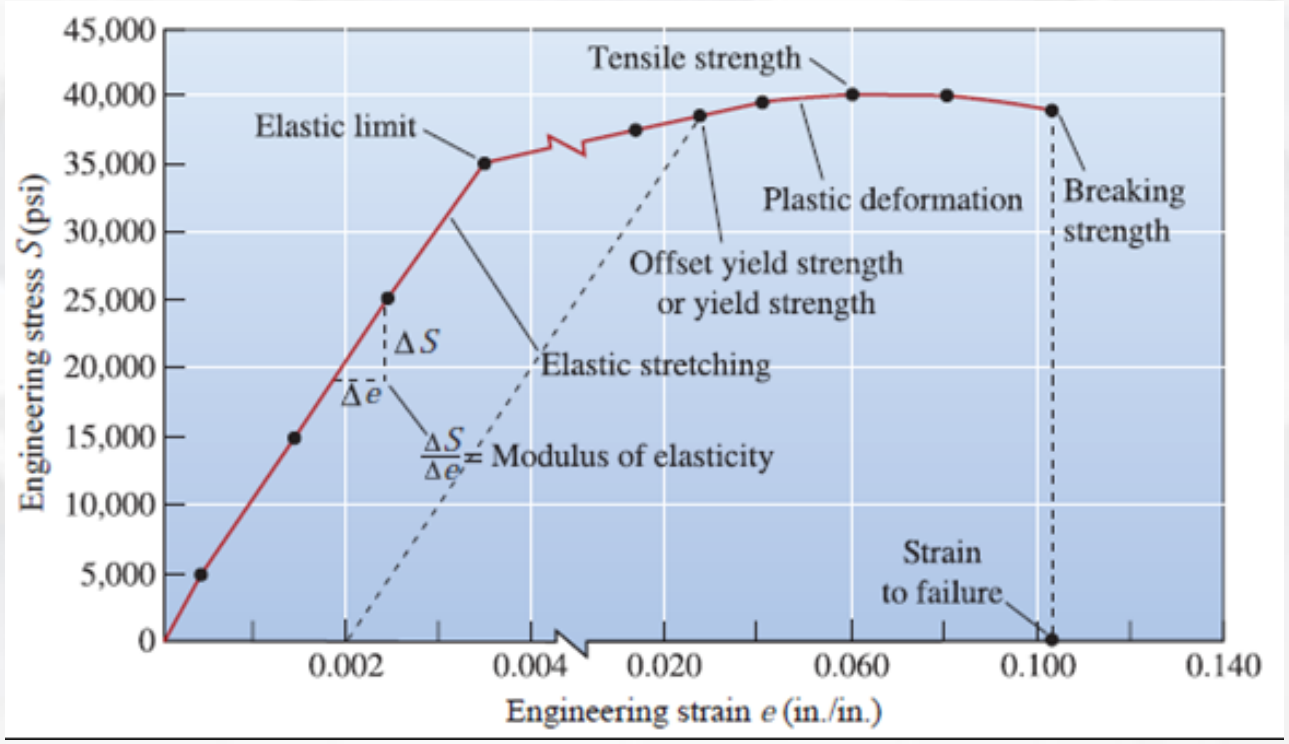
L_0 Original distance between the gauge marks before the test is done

l Distance between the gauge marks after the force F is applied

A_0 Original cross-sectional area before the test

Assignment

An aluminium rod is to withstand an applied tensile force of 300 kN. The maximum allowable stress on the rod is fixed at 170 x 10^3 kNm^{-2}, to ensure sufficient factor of safety. If the rod should be at least 4 m long, but must not deform more than 8 mm when the force is applied, design a rod for this purpose.



Example 2-7 Continues

Solution

Under normal circumstances, the expected cross-sectional area:

$$A_0 = F/\sigma = (300)/(170x10^3) = 1.7647x10^{-3}\ m^2 = 1764.7\ mm^2$$

The rod should have a cross-sectional area of 1765 mm^2.

For a cylindrical rod, $d = \sqrt{A_0/\pi} = \sqrt{1764.7/\pi} = 47.4\ mm$

From Fig. E3-7, the strain expected for the stress 170 x 10^3 kNm^{-2} is 0.0025.
For the cross-sectional area above, the length of the rod is:

$$\varepsilon = \Delta l/l = 8/l \Rightarrow l = 8/\varepsilon = 8/0.0025 = 3200\ mm = 3.2\ m\ (< 4\ m)$$

Example 2-7 Continues

The minimum length of the rod is given as 4 m.

To keep this, the cross-sectional area should be larger

$$\varepsilon = \Delta l / l = 8 / 4000 = 0.002$$

From Fig. E3-7, the corresponding stress for the strain of 0.002 is 140 x 10^3 kNm^{-2} which is less than the given maximum of 170 x 10^3 kNm^{-2}.

The minimum cross-sectional area then, is

$$A_0 = F / \sigma = (300) / (140x10^3) = 2.143x10^{-3}\ m^2 = 2143\ mm^2$$

Example 2-7 Continues

To satisfy, the conditions, maximum stress of 170 x 10^3 kNm^{-2} and maximum elongation of 8 mm: $A_0 = 2143\ mm^2$.

Therefore the Diameter, d is $d = \sqrt{A_0/\pi} = \sqrt{2143/\pi} = 52.24\ mm$

Thank You